

Xinbao Qiao

xinbaoqiao@cuhk.edu.hk | [Homepage](#) | [GitHub](#) | [Google Scholar](#)

EDUCATION

The Chinese University of Hong Kong
PhD student in Information Engineering

Hong Kong, China
2026–present

Zhejiang University
M.Sc. in Artificial Intelligence

Hangzhou, China
2022–2025

- Major GPA: 90/100; rank: 3/25.
- Selected coursework: AI Algorithms and Systems (100), Secure Artificial Intelligence (97).

Shandong University
B.Eng. in Communication Engineering

Jinan, China
2018–2022

- Major GPA: 82.47/100.
- Third-class Academic Award, 2018–2019.

RESEARCH INTERESTS

Data-centric machine learning; AI for Networks; Networks for AI; trustworthy AI; machine unlearning; synthetic-data evaluation and model collapse.

PUBLICATIONS

- [1] **When Sample Selection Bias Precipitates Model Collapse.**
Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang. *ICML, 2026*.
- [2] **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness.**
Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang. *AAAI, 2026*.
- [3] **Hessian-Free Online Certified Unlearning.**
Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei. *ICLR, 2025*.
- [4] **DynFrs: An Efficient Framework for Machine Unlearning in Random Forest.**
Shurong Wang, Zhuoyang Shen, Xinbao Qiao, Tongning Zhang, Meng Zhang. *ICLR, 2025*.
- [5] **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification for Pretrained Model.**
Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang. Under review, *NeurIPS 2026*.
- [6] **Illusory Pattern Perception Drives Spurious Inference in Large Language Models.**
Peihua Mai, Zhuoyan Shao, Xinbao Qiao, Meng Zhang, Xinyue Zhou, Yan Pang. Under review, *NeurIPS 2026*.
- [7] **Learn What Matters: Data Pruning for Efficient Decentralized Learning.**
Xinbao Qiao, Xunhao Jiang, Zuozhu Liu, Peng Sun, Meng Zhang. Under review.

RESEARCH EXPERIENCE

The Chinese University of Hong Kong
PhD student and Research Assistant

Hong Kong, China
2026–present

- Study data-centric machine learning, AI for Networks, and Networks for AI under decentralized data access.
- Current focus: communication-aware learning, reliable evaluation, and distributed methods for collaborative distributional references.

National University of Singapore Research Institute (Chongqing)
Full-time Research Intern; advisor: Yan Pang

Chongqing, China
2025

- Worked on trustworthy LLM systems and synthetic-data evaluation.
- Developed collaborative Wasserstein-geometry methods for evaluating recursive synthetic-data training under low-resource verification and local sample-selection bias.

- Studied data influence attribution, certified data removal, and machine unlearning.
- Developed methods that examine trade-offs among fairness, robustness, privacy, and utility in data-centric learning systems.

SELECTED ACADEMIC PROJECTS

- **AI and networks:** study distributed computation and collaborative evaluation for learning systems whose data and evidence are spread across networked parties.
- **Synthetic-data model collapse:** analyzed how locally biased sample selection can accelerate model collapse in low-resource communities, and proposed collaborative Wasserstein proxy references for distributed evaluation.
- **Machine unlearning:** designed Hessian-free certified unlearning and soft-weighted unlearning methods for privacy, fairness, and robustness settings.

SKILLS

Programming: Python, PyTorch, TensorFlow, MATLAB, C/C++, SQL, Linux, Docker, Git, $\text{\LaTeX}$.

Languages: Mandarin Chinese (native), English.